AGRICULTURAL DRAINAGE · NOTE DR05

Tile Drain Design — Depth and Spacing, Leaching Requirement, Hooghoudt Equation and Drain Sizing

Course topics: IDE-502 T6, IDE-502 T7, IDE-502 T8, IDE-502 T11, IDE-502 T12, IDE-502 T13, IDE-501 T7

What the design decides

Tile drainage design selects the depth and spacing of the tiles and the related hydraulic and soil factors. It is governed by the soil's hydraulic conductivity, the infiltration rate during irrigation, the optimum water-table depth, the crop's response to a fluctuating water table, the depth of impermeable strata, the outlet depth and trenching limits, and crop needs. Earlier practice relied on experience and judgement.

Soil (Table 11.1)	Spacing (m)	Depth (m)
Clay	9–15	0.90–1.05
Clay loam	12–21	0.90–1.05
Loam	18–30	1.05–1.20
Sandy loam	30–60	1.20–1.50

- Minimum depth to the tile bottom ≈ 60 cm + the tile diameter; in uniform permeable soils the depth is usually 0.75–1.50 m.
- **Basic design factors:** hydraulic conductivity, infiltration rate, percolation rate (for standing water, as in paddy fields) and leaching requirement.

Leaching requirement

Leaching removes soluble salts by passing water through the soil. The **leaching requirement (LR)** is the extra water, beyond the crop water need, that must pass through the root zone to keep salinity in check. Find it from climate data and the salt concentrations of the soil and the irrigation water.

$$LR = (E - P)\cdot C_i \,/\, [F\cdot(C_fc - C_i)]$$

LR = leaching requirement (mm); E = evapotranspiration (mm); P = effective rainfall (mm); C_i = salt concentration of irrigation water (me/l); C_fc = salt concentration at field capacity (me/l); F = leaching efficiency (fraction)

- If P > E, LR is negative — no drainage is needed for leaching.
- The higher the root-zone salinity the crop can tolerate (higher C_fc), the lower the LR.
- Check (salt balance): salt in with the water applied, $(E - P + LR)\cdot C_i$, must equal salt out, $F\cdot LR\cdot C_fc + (1 - F)\cdot LR\cdot C_i$; solving gives the formula above.

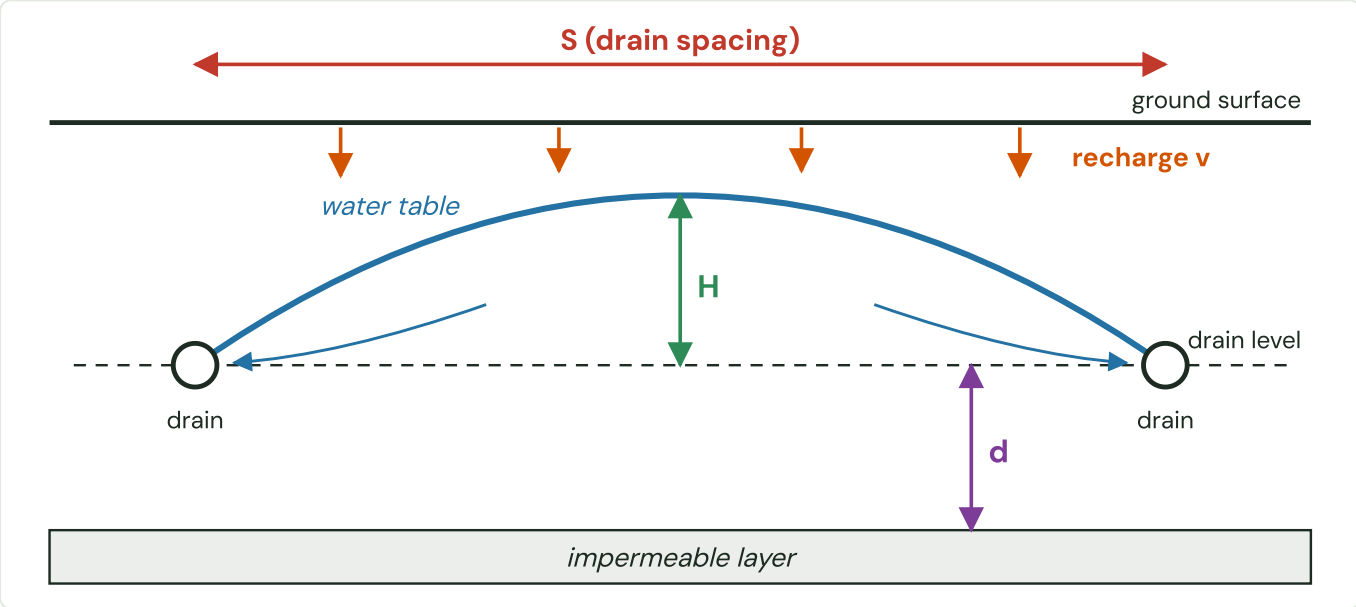
Illustration (not in the notes)
E − P = 600 mm, C_i = 2 me/l, C_fc = 10 me/l, F = 0.8
LR = 600 × 2 / [0.8 × (10 − 2)] = 1200 / 6.4 = 187.5 mm of extra water

Hydraulic design

- Base the water-removal requirement on excess irrigation water, rainfall beyond the soil's storage, or the leaching requirement of salt-affected soils — whichever is most critical.
- **Steady-state approach:** the removal rate equals the drainage coefficient, with an equilibrium water table.
- **Non-steady-state approach:** the water table fluctuates; design for the rate of fall needed to protect crops. Often better for irrigated or tropical conditions, though steady-state results may still be acceptable.

Hooghoudt's equation for drain spacing

- **Assumptions:** homogeneous soil of conductivity k; drains evenly spaced at S; hydraulic gradient = slope of the water table (dy/dx); Darcy's law holds; impermeable layer at depth d below the drains; recharge from rain or irrigation at rate v; origin on the impermeable layer below the drain centre.
- A lower layer whose permeability is ≤ one-tenth of the upper layer's can be treated as impermeable.



Hooghoudt's geometry: drains at spacing S, water table H above drain level midway between them, impermeable layer d below the drains, uniform recharge v.

Flow across a vertical plane at x: $q_x = (S/2 - x)\cdot v = k\cdot y\cdot dy/dx$

Integrated: $S^2 = 4k(H^2 - h^2 + 2dH - 2dh)/v$

Empty drain ($h = 0$): $S^2 = 4kH(2d + H)/v$

Water-table shape (ellipse): $y^2/(S^2v/4k) + x^2/(S^2/4) = 1$

H = height of the water table midway between drains, and h = water level in the drain, both above drain level. (Integrating from y = d + h at the drain to y = d + H midway gives $v\cdot S^2/8 = (k/2)[(d + H)^2 - (d + h)^2]$, which expands to the equation above.)

Illustration (not in the notes)
k = 1.0 m/day, v = 2 mm/day = 0.002 m/day, H = 0.5 m, d = 2 m, empty drains
$S^2 = 4 \times 1.0 \times 0.5 \times (2 \times 2 + 0.5) / 0.002 = 4500$ m² → S ≈ 67 m

Drainage coefficient for tile drains

- The depth of water to remove in 24 hours; in humid regions it depends mainly on rainfall.
- Annual rainfall 750–1500 mm → 10–13 mm/day. Increase by 30–50% for organic soils or high-value crops. If tiles also carry surface runoff → 20–25 mm/day.
- In irrigated areas drain discharge is about 10–50% of the water applied, depending on irrigation depth, application method, leaching requirement and soil.

US Soil Conservation Service: $D.C. = (P + c) \cdot i / F$

P = deep-percolation fraction; c = seepage-loss fraction; i = irrigation depth (mm); F = irrigation interval (days)

Correction: the notes print D.C. = (P + c)/(iF). A unit check shows the irrigation depth must multiply: the fractions (P + c) of a depth i (mm) lost every F days give a removal rate in mm/day.

Capacity and size of tile drains

- Size tiles with Manning's equation, keeping velocity ≥ about 0.4 m/s to avoid silting.
- **Minimum slopes:** 0.1% for 10 cm tile; 0.05% for 15 cm tile.
- **Manning's n:** 0.013 for clay and concrete pipe; 0.015–0.02 for corrugated plastic tubing.
- Choose the lateral diameter for the flow at its downstream end; about 10 cm is the common minimum. Collectors may reach 75 cm. Base capacity on the whole drainage area, not just the last tile; compute sizes only for laterals longer than 150 m.
- **Joints:** water enters through the cracks between tiles; doubling the crack width raises inflow only ~10%. Use 1–3 mm cracks in silt loam and silty clay loam; in sandy soils butt the ends as close as possible. Laboratory inflow-meter tests fix the spacing.
- **Perforations:** each extra hole adds relatively less flow; larger holes admit more; the effect is smaller at greater depth; envelope materials can reduce the number of perforations needed.

Selecting tiles

- Use first-grade tiles only: crack-free, circular and straight, smooth inside, with smooth regular ends for joints ≤ 3 mm, and a clear ring when tapped.
- Length: about 30 cm for diameters under 30 cm; up to 75 cm for larger diameters.
- **Clay tiles:** common or vitrified. Vitrified tiles are stronger and less porous but costly, so they are rarely used in agriculture.
- **Concrete tiles:** cement : sand = 1 : 3–4. Suitable in non-acid, non-alkali soils; use special cement in acid or alkali soils.

Sources: ICAR eCourse 'Drainage Engineering' — Dr M. K. Jha (IIT Kharagpur), reviewed by Prof. K. Yellareddy (Director A&R, WALAMTARI, Hyderabad); AgriMoon e-book. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Hooghoudt's spacing equation for an empty drain is

(a) $S^2 = 4kH(2d + H)/v$ (b) $S = 4kH/v$

(c) $S^2 = 8kdH/v$ (d) $S^2 = kH^2/v$

Q2. A lower soil layer can be treated as impermeable if its permeability is at most

(a) half
(b) one-fifth
(c) one-tenth
(d) one-hundredth

Q3. In the leaching-requirement formula, if effective rainfall exceeds ET then LR is

(a) maximum
(b) negative — no drainage needed for leaching
(c) equal to ET
(d) infinite

Q4. Tile drainage coefficient for 750–1500 mm annual rainfall is about

(a) 2–4 mm/day
(b) 10–13 mm/day
(c) 20–25 mm/day
(d) 50 mm/day

Q5. Manning's n for clay and concrete tile is

(a) 0.010
(b) 0.013
(c) 0.020
(d) 0.035

Q6. Minimum velocity in a tile drain to avoid silting is about

(a) 0.1 m/s
(b) 0.4 m/s
(c) 1.0 m/s
(d) 2.0 m/s

Q7. Doubling the crack width between tiles increases inflow by about

(a) 10%
(b) 50%
(c) 100%
(d) 200%

Q8. Concrete tiles are made with a cement : sand ratio of

(a) 1 : 1
(b) 1 : 2
(c) 1 : 3–4
(d) 1 : 8

Q9. Typical tile spacing in sandy loam (Table 11.1) is

(a) 9–15 m
(b) 12–21 m
(c) 18–30 m
(d) 30–60 m

Answers

Q1 — (a) $S^2 = 4kH(2d + H)/v$
Q2 — (c) one-tenth — *of the upper layer.*
Q3 — (b) negative — no drainage needed for leaching
Q4 — (b) 10–13 mm/day
Q5 — (b) 0.013
Q6 — (b) 0.4 m/s
Q7 — (a) 10%
Q8 — (c) 1 : 3–4
Q9 — (d) 30–60 m